

Measuring Retrieval

A Practical Technical Book on IR and RAG Metrics

Draft production edition

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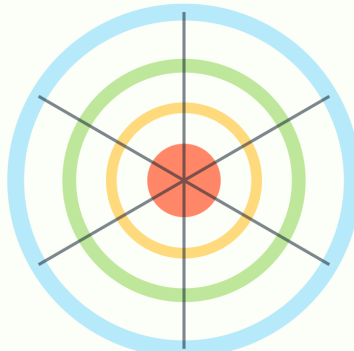
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1 Measuring Retrieval

This book gives a compact working theory of retrieval evaluation. It is written for teams that build search, recommendation, and retrieval augmented generation systems.

The goal is operational clarity. A metric should tell a team what decision to make, what risk remains, and what companion measurement belongs beside it.

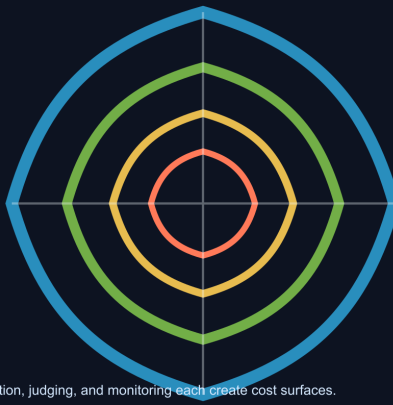
Frontend Fractal: What The User Sees



Answer quality branches into clarity, citation, latency, and visible confidence.

Figure 1: Light frontend fractal

Backend Fractal: What The System Pays



Indexing, retrieval, generation, judging, and monitoring each create cost surfaces.

Figure 2: Dark backend fractal

2 1. The Four Questions

Retrieval evaluation starts with four questions.

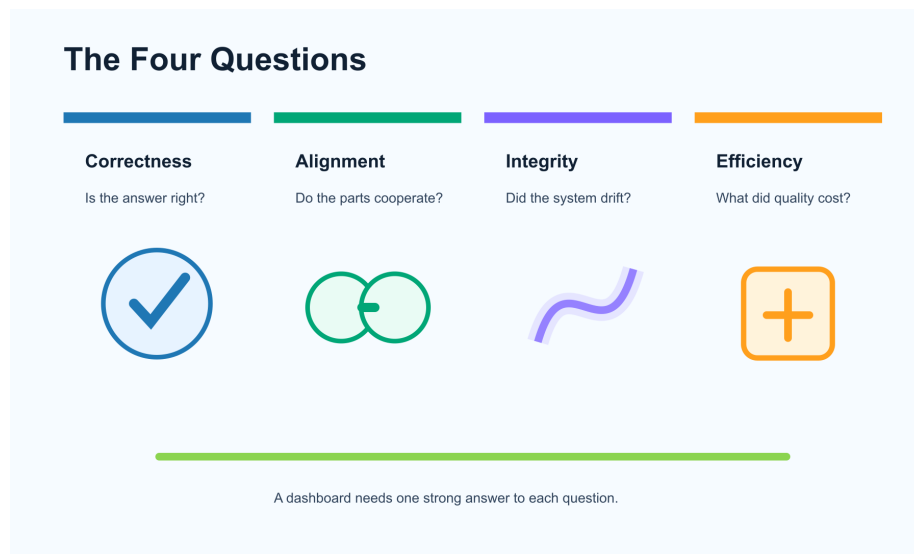


Figure 3: The four questions

Dimension	Question	Typical failure it catches
Correctness	Is the retrieved or generated content right?	Irrelevant result, unsupported answer, wrong citation
Alignment	Do the retriever and generator cooperate?	The answer is correct while retrieval contributed little
Integrity	Has the system changed over time?	Embedder drift, corpus drift, judge drift
Efficiency	What did quality cost?	Latency, token waste, redundant context, rebuild expense

Every dashboard should answer all four questions. Many teams answer the first question many times.

Correctness is the local question. It asks whether a document, claim, answer, or citation passes a judgment.

Alignment is the system question. It asks whether the retrieved evidence helped the generator produce the answer.

Integrity is the time question. It asks whether the score has the same meaning after a release, model change, corpus update, or judge change.

Efficiency is the resource question. It asks whether the extra quality is worth the latency, tokens, engineering work, and operational risk.

3 2. Facets

A metric needs a label before it needs a number. The label has four facets.

Facet	Values	Use
Unit	document, passage, claim, answer, session, system	Tells what item receives a score
Supervision	judged, reference-based, reference-free, behavioral	Tells where truth comes from
Stage	retrieval, generation, citation, evaluation, deployment	Tells which layer owns the fix
Tier	diagnostic, release gate, monitor, audit	Tells how often to compute it

Write the facet line under every metric name. A reader should know where the number lives before seeing the formula.

Two metrics can share a formula shape and answer different product questions. Precision over retrieved documents, citation precision over cited passages, and claim precision over generated assertions belong to different operational layers.

The unit matters most. A claim-level score can improve when answers become shorter. A document-level score can improve while citations stay wrong. A session score can hide one catastrophic answer inside a long interaction.

The stage matters next. Retrieval teams can act on nDCG, recall, and redundancy. Generator teams can act on claim support and citation faithfulness. Platform teams can act on latency, token cost, and judge stability.

4 3. Signals Need Decisions

A signal is an observation. A metric is a decision instrument.

Mean embedding distance is a signal. Token count is a signal. Number of retrieved chunks is a signal. They become metrics when paired with a defined decision, threshold, or comparison.

Use this test: if the number moves, can the team name the action? If yes, it is ready for the dashboard. If no, keep it in the analysis notebook.

Common signals in retrieval work:

Signal	Useful action
Retrieved chunk count	Debug fanout and routing
Mean similarity	Inspect retriever confidence
Token count	Estimate cost
Query length	Segment traffic
Judge confidence	Prioritize human review

Production metrics should be boring. They should send engineers to the correct subsystem.

5 4. Classical IR Metrics

Classical IR gives the measurement grammar for retrieval. RAG changes the downstream system. The same grammar still matters.

5.1 Set metrics

Precision measures the share of retrieved items that are relevant. Recall measures the share of relevant items that were retrieved. F1 combines the two when a single operating point is useful.

Metric	Use	Watch
Precision	Topical cleanliness	Can reward timid retrieval
Recall	Coverage	Needs a known relevant set
F beta	Product-weighted tradeoff	Hides the separate movement of precision and recall
Precision@k	Top-k quality	Ignores relevant items below k

5.2 Rank metrics

Rank metrics add position. They model attention. High ranks matter more because users and generators consume early items first.

Metric	User model	Best fit
MRR	First useful hit ends the search	Known-item lookup
MAP	Every relevant item matters	Recall-rich search
nDCG	Graded relevance with rank discount	General ranked retrieval
ERR	User may stop after satisfaction	Navigational search
RBP	Persistence at each rank	Stable, interpretable dashboards

For RAG, compute rank metrics at the injected depth. The generator can use only the context it receives.

5.3 Incomplete judgments

Judgment pools are incomplete in real systems. `bpref` and `infAP` reduce sensitivity to missing labels. They are useful when old test collections meet new retrievers.

5.4 Comparing rankings

Kendall tau and Spearman rho compare full ranked lists. RBO compares top-heavy overlap and works with unequal lists. RBO is the practical drift metric for monthly retrieval monitoring.

5.5 Metric cards

5.5.1 Precision

Formula: retrieved relevant items divided by retrieved items.

Use it when the product experience is damaged by junk results. Legal search, clinical lookup, and support automation often need high precision near the top of the list.

Read precision with recall. High precision can come from a cautious retriever that returns too little.

5.5.2 Recall

Formula: retrieved relevant items divided by all relevant items.

Use it when missing evidence is the primary harm. Compliance review, patent search, literature review, and retrieval for synthesis belong here.

Recall needs a denominator. In live systems that denominator is usually approximated by pooled judgments, curated cases, or task-specific audits.

5.5.3 F beta

Formula: weighted harmonic mean of precision and recall.

Use beta greater than 1 when misses are costly. Use beta less than 1 when junk results are costly. Report the chosen beta beside the score.

5.5.4 MRR

Formula: average reciprocal rank of the first relevant result.

Use it for known-item tasks: password reset article, account policy page, exact entity lookup. The first useful hit ends the search.

5.5.5 MAP

Formula: average of precision values at relevant ranks, then averaged across queries.

Use it when many relevant items matter. MAP rewards systems that retrieve relevant items early and consistently.

5.5.6 nDCG

Formula: discounted cumulative gain divided by ideal discounted cumulative gain.

Use it when judgments are graded. nDCG handles “excellent”, “acceptable”, and “weak” relevance in one score.

For RAG, choose k as injected context depth. If the retriever returns 50 passages and the generator receives 8, nDCG@8 describes the usable ranked evidence.

5.5.7 RBP

Formula: rank-biased precision with persistence parameter p.

Use it when you want a stable top-heavy score. The parameter p expresses how likely the user or downstream consumer is to continue to the next rank.

5.5.8 RBO

Formula: rank-biased overlap between two ranked lists.

Use it for drift. Compare today’s ranking with a frozen baseline at the same k. Keep the query set and corpus timestamp visible.

6 5. RAG Metrics

RAG evaluation has three objects: evidence, answer, and judge.

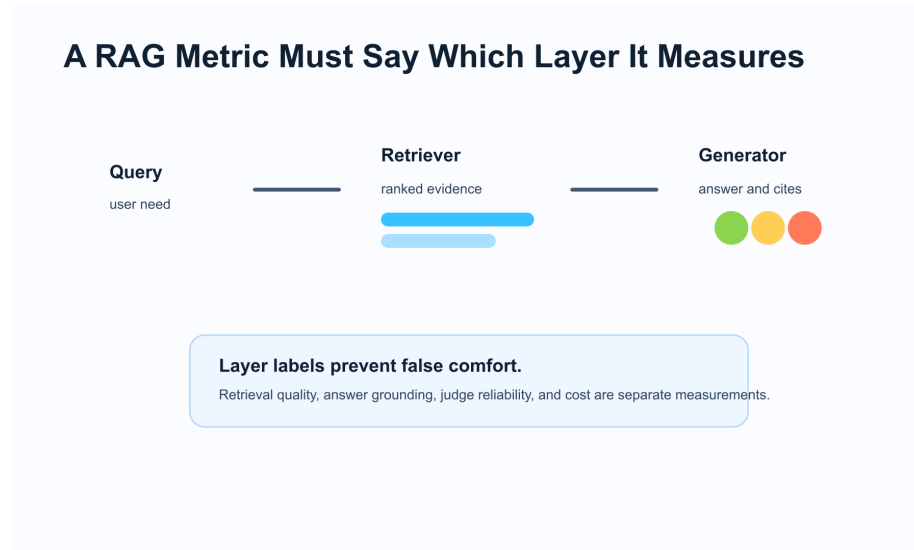


Figure 4: RAG layer diagram

6.1 Correctness and grounding

Claim decomposition is the clean unit for answer evaluation. Break the answer into atomic claims. Score whether each claim is supported, contradicted, or absent in the retrieved context.

RAGChecker is valuable because it separates retriever failures from generator failures. Claim Recall estimates whether required claims were found. Self-Knowledge estimates the rate of answers supplied from model memory. Hallucination estimates unsupported generated content.

RAGAS is useful for reference-free evaluation. Treat paper definitions and library implementations as separate instruments. Pin the version.

ALCE-style citation metrics ask whether citations support the sentence they cite. The leave-one-out trick tests whether a citation is doing real work.

6.2 Alignment

Alignment asks whether retrieval helped generation. Contribution metrics estimate the utility of a document or passage by measuring what changes when it is present, absent, or perturbed.

Family	Measures	Use
eRAG / DIG / SePer / Gain	Passage contribution to answer quality	Diagnose context utility

Family	Measures	Use
WARG	Gap between retrieved relevance and generated attribution	Detect wasted retrieval and low-rank dependence
CUE / MIRAGE quadrants	Relationship between answer quality and context use	Find lucky guesses and blind spots
UDCG	Utility and distraction in ranked context	Tune retrieval for generator use

Pick one contribution metric for the dashboard. Use the others as audit tools during method work.

6.3 Integrity

Integrity metrics watch change. Use RBO against a frozen baseline for rank drift. Use version-sensitive question sets for corpus drift. Use anchor sets and Krippendorff alpha for judge drift.

6.4 Efficiency

Efficiency should include index-time, query-time, token-time, and evaluation-time costs.

Cost center	Metric
Indexing	rebuild time, index size, embedding cost
Query serving	TTFT p50, TTFT p95, total latency
Context	tokens per query, pairwise redundancy, evidence hit rate
Evaluation	judge calls, human review minutes, audit cost

6.5 Metric cards

6.5.1 Claim Recall

Unit: claim.

Question: did the retrieved context contain the information needed for the answer?

Use it as the ceiling metric. A generator cannot ground a claim in evidence that never arrived. Low Claim Recall sends work to retrieval, query rewriting, corpus coverage, or chunking.

6.5.2 Self-Knowledge

Unit: answer or claim.

Question: how often did the model answer correctly from parametric memory while retrieved evidence missed the answer?

Use it to separate product success from retrieval success. This metric matters in domains where model memory is strong enough to hide retrieval failure.

6.5.3 Hallucination

Unit: claim.

Question: which generated claims lack support?

Use it as a release gate for high-risk surfaces. Pair it with average claim count, because short answers generate fewer opportunities for unsupported claims.

6.5.4 Citation Precision

Unit: citation.

Question: does the cited source support the sentence that cites it?

Use it when users see citations. A citation should carry evidence. Sentence-level checking gives the cleanest product signal.

6.5.5 Contribution metrics

Unit: passage or document.

Question: how much did this retrieved item improve the generated answer?

Use one member of the family for the dashboard. Use perturbation studies and ablations during method development.

6.5.6 WARG

Unit: query.

Question: does generated attribution follow retrieved relevance?

Use it before investing heavily in retriever ranking improvements. A low alignment score means the generator may be drawing value from low-ranked passages or ignoring top-ranked passages.

6.5.7 Judge agreement

Unit: labeled item.

Question: does the automated judge agree with the human anchor set?

Use Krippendorff alpha for multi-label or missing-label settings. Report the anchor set date and judge snapshot.

7 6. The Starter Dashboard

Start with eight measurements and two context lines.

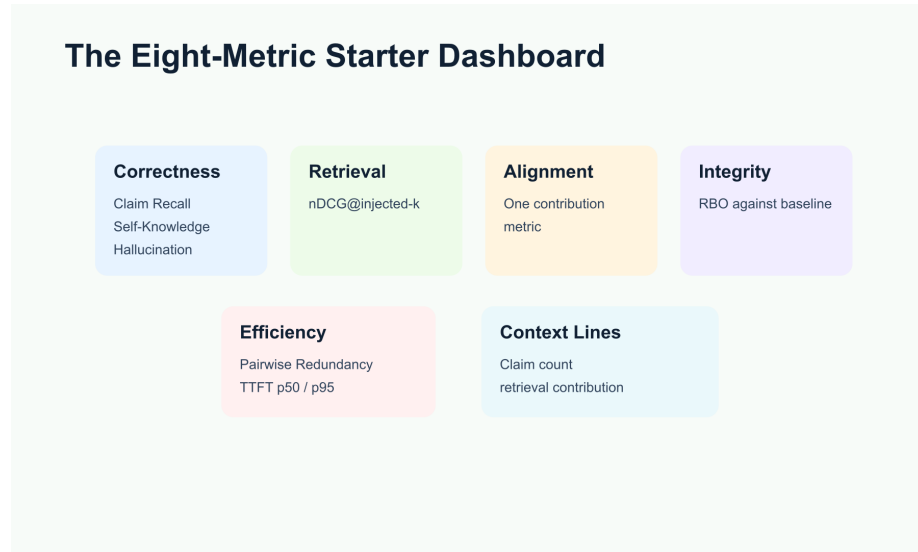


Figure 5: Core dashboard

Number	Metric	Dimension	Reason
1	Claim Recall	Correctness	Sets the ceiling for grounded answer quality
2	Self-Knowledge	Correctness	Measures correct answers supplied without retrieved support
3	Hallucination	Correctness	Tracks unsupported generated content
4	nDCG@injected-k	Correctness	Keeps continuity with retrieval evaluation

Number	Metric	Dimension	Reason
5	One contribution metric	Alignment	Measures whether context helps the answer
6	RBO against frozen baseline	Integrity	Detects rank drift cheaply
7	Pairwise Redundancy	Efficiency	Finds repeated context and wasted tokens
8	TTFT p50 / p95	Efficiency	Measures the latency users feel first

Context lines:

Line	Reason
Average claim count per answer	Prevents brevity from looking like quality
retrieval contribution experiment	Shows whether retrieval earns its operational cost

Report each metric with owner, unit, denominator, evaluation set, judge version, and date.

7.1 Dashboard rows

Each row should include:

Field	Purpose
Metric	Names the instrument
Dimension	Names the question
Owner	Names the team that can act
Unit	Prevents denominator confusion
Set	Names the evaluation slice
Version	Pins corpus, embedder, generator, and judge
Last run	Makes staleness visible
Action band	Converts movement into work

7.2 Action bands

Action bands make a score operational.

Band	Meaning	Action
Green	Within expected range	Monitor
Yellow	Movement needs review	Sample failures
Red	Product risk	Block release or roll back

Use bands sparingly. A band with no owner creates noise.

8 7. A 90-Day Rollout

8.1 Week 1

Freeze the baseline. Save 500 queries, top-k results, embedder version, retriever version, corpus timestamp, injected k, and judge version.

Create a 100-item human-labeled anchor set. Time a full index rebuild. Record the number.

8.2 Weeks 2 to 4

Run the retrieval contribution experiment. Measure semantic test coverage. Compute pairwise redundancy. Compare MAP and bpref to estimate judgment distortion. Run a position-bias test on the judge.

8.3 Month 2

Run a k sweep over $k = 1, 3, 5, 10, 20, 50$. Plot recall, answer quality, redundancy, tokens, and TTFT. Set k at the knee of the curve.

8.4 Month 3

Run the full diagnostic layer offline. Promote three RAGChecker metrics. Sample CUE quadrants. Run WARG once to test whether the generator follows the retrieved ranking. Build a 50-question version-sensitive set.

8.5 Ongoing

Run RBO monthly. Run expensive diagnostics per release. Run the judge anchor set for every judge change. Run Jaccard and RankSimilarity for every embedder change.

9 8. Design Review Rules

9.1 Rule 1. Faithfulness travels with retrieval hit rate

Faithfulness scores generated support. Retrieval hit rate scores whether the required evidence arrived. A team needs both numbers on one row.

9.2 Rule 2. Relevance labels give partial evidence of generator utility

A passage can be topically relevant and harmful to reasoning. A passage can be tangential and useful. Use a contribution metric when generation is the product surface.

9.3 Rule 3. Judge changes are system changes

A new judge can move scores while the product stays fixed. Pin the judge. Re-score the anchor set. Report alpha.

9.4 Rule 4. Cost belongs in the same review as quality

Latency and tokens shape the feasible system. The cost-quality frontier is the artifact that lets a team choose a point deliberately.

9.5 Rule 5. The metric should name the owner

Retriever, generator, evaluation, platform, and product teams need separate accountability. A metric that names no owner becomes ceremony.

9.6 Rule 6. Name the denominator

A rate without its denominator invites misreadings. Claim rate, answer rate, query rate, and session rate can all move differently.

9.7 Rule 7. Keep the failure sample

Every release review should include sampled failures beside aggregate scores. The sample teaches the team what the number means this week.

9.8 Rule 8. Use expensive metrics as diagnostics

Some metrics are too costly for every build. Run them during release review, incident review, and method evaluation. Promote only the cheap, stable, high-signal subset to continuous monitoring.

10 Build Notes

The book is Pandoc-ready Markdown.

Build a PDF from the `measuring-retrieval-book` directory:

```
pandoc metadata.yaml chapters/*.md --css assets/book.css --toc --number-sections -o measuring-retrieval-book.pdf
```

Build HTML for quick review:

```
pandoc metadata.yaml chapters/*.md --css assets/book.css --toc --number-sections -s -o measuring-retrieval-book.html
```

The SVG diagrams are in `assets/diagrams`. The frontend and backend fractals are in `assets`.

11 Appendix A. Metric Selection Table

Product question	First metric	Companion metric	Owner
Are top results clean?	Precision@k	Recall@k	Retrieval
Did we find enough evidence?	Claim Recall	nDCG@injected-k	Retrieval
Are answers grounded?	Hallucination rate	Average claim count	Generation
Are citations useful?	Citation Precision	Citation Recall	Generation
Is the generator using retrieval?	Contribution metric	WARG	System
Did retrieval drift?	RBO against baseline	Jaccard@k	Retrieval
Did the judge drift?	Anchor-set alpha	Score delta by slice	Evaluation
Are we wasting context?	Pairwise Redundancy	Tokens per answer	Platform
Is latency acceptable?	TTFT p95	Total latency p95	Platform
Is evaluation affordable?	Judge calls per release	Human review minutes	Evaluation

11.1 Minimal release report

1. One paragraph describing the system change.
2. One table with the eight starter metrics.
3. One table with changed versions.
4. Ten sampled failures.

5. One decision: ship, hold, roll back, or run a targeted audit.

12 Appendix B. PDF Production Checklist

12.1 Before building

Run a structure scan:

```
rg -n "^#|^##|^###" chapters
```

12.2 Build

```
make pdf
```

12.3 Review

Check the cover, table of contents, diagrams, tables, and page breaks.

12.4 Suggested next production pass

1. Replace placeholder author metadata.
2. Add citation keys if the PDF needs formal references.
3. Add chapter openers from the original whitepaper source where more narrative depth is wanted.
4. Add a glossary for teams new to IR.